



Swinburne University of Technology Hawthorn Campus Department of Computing Technologies

COS20028 Big Data Architecture and Application Lab Task 5 - Semester 2

Assessment Title: Analysing Ad campaign data with Pig.

Assessment Outcome: Pass/Fail (for practice only)

Assessable Item:

- One (1) piece of a report containing the complete Apache Pig code for preparing the data and finding the analytic result, with the screenshot of submitted commands and execution results.

Purpose of Assignment

This assignment evaluates whether a student understands how to achieve an analytic goal via Apache Pig.

This is an individual assignment and is mandatory for passing this unit.

Introduction

The source data is built in the virtual machine image under the path of **/home/training/training_materials/analyst/data**. Our goal is using Apache Pig to find out the **top five (5)** websites that associated with the high cost in **France** from "**ad_data1.txt**" and in **any country** from "**ad_data2.txt**". The cost for putting up an advertisement is recorded in the *cpc* attribute. Whether an advertisement is clicked is recorded by "**1**" in the *was_clicked* attribute.

The samples and the corresponding attribute name are listed as follows:

ad_data1.txt:

Keyword	Champaign_id	Date	Time	Display_Site
lightweight	D8	05/01/2013	00:00:08	gamersite.example.com
Was_Clicked	CPC	Country	Placement	
0	72	USA	SIDE	

- The above example tuple shows that this advertisement is in the D8 champaign with the keyword "lightweight", displayed in USA on the website "gamersite.example.com" at 00:00:08 on 05/01/2013 on the side of the webpage. The user didn't click on this advertisement. Otherwise, the company will be charged 72 cents.

ad_data2.txt:

Campaign_id	Date	Time	Display_Site
A1	05-01-2013	00:00:47	techwiz.example.com
Placement	Was_Clicked	CPC	Keyword
SIDE	0	62	camera

- The above example tuple shows that this advertisement is in the A1 campaign with the keyword “camera”, displayed on the website “techwiz.example.com” at 00:00:47 on 05/01/2013 on the side of the webpage. The user didn’t click on this advertisement. Otherwise, the company will be charged 62 cents.

Assignment Goal

This assignment aims at training students to get familiar with Apache Pig for managing data into the specific format for further uses.

Assignment Task

The following tasks should be accomplished:

1. Write a set of Apache Pig codes for organising the data from multiple sources into the identical format.
2. Write an Apache Pig code and use file patterns (globs) to load files for analysing the content to find the sum of cost based on the value in the *cpc* attribute. Note that only the advertisement, which is clicked will be charged and have the corresponding cost. If the advertisement is not clicked, it should not be included in the calculation.

Please use the template to prepare your submission.

Submission Requirement

Only the submission using the given template will be evaluated by the tutors.

Failure to adhere to the submission requirements will immediately result in a failure for this assignment.

Rubric: Lab Task 5

Task	Assessment Outcome
Successfully finding the answer to the given question in the template and the submitted code in the report is reasonable, complete, and fully functional.	Pass/Fail

----- End of Assignment -----